Mechanism checks (research period only)

1. Autocorrelation of overnight and intraday returns (Nifty)

lag	overnight (gap)	intraday	close-to-close	corr(gap_t, intraday_t+lag)	corr(intraday_t, gap_t+lag)
1	-0.034	-0.051	-0.001	0.057	0.038
2	0.123	-0.063	0.001	0.026	-0.018
3	0.025	-0.012	0.005	-0.014	0.025
4	-0.036	0.036	-0.019	-0.07	0.003
5	0.153	-0.019	0.089	0.02	0.088

same-day correlation of gap and intraday: -0.029

period	gap lag1	gap lag2	close-to-close lag1
2012-2016	0.034	0.04	0.077
2017-2021	-0.073	0.164	-0.055
excl. 2020	0.021	0.02	0.076

2. next tradable day on several lags at once (Nifty)

	coef	t
const	0	0.503
gap0	0.217	3.906
gap1	0.07	0.686
gap2	-0.074	-0.64
intra0	-0.063	-2.054
intra1	-0.05	-0.908

3. Same trading rule, different signal (k=0.5, w=60, h=2, long/short, costs 0.05% a side)

signal	net Sharpe	CAGR %	trades	time in market %
overnight gap (the real rule)	1.25	20.19	661	77.7
close-to-close return	-0.28	-5.54	756	78.44
intraday return (open to close)	-1.14	-17.91	801	78.97
gap minus intraday	0.7	10.55	786	78.89
overnight gap of BANK NIFTY	0.98	15.18	687	72.67
pure noise (shuffled gaps)	-0.45	-7.82	613	69.03

4. Baselines (Nifty, open to open, same costs)

baseline	net Sharpe	CAGR %	MaxDD %
always long	0.85	14.4	-37.77
long 78% of days at random (no signal)	0.7	10.49	-34.87
long only after an up close (momentum, 1 day)	0.19	1.6	-30.63
long only after an up gap (no threshold, 1 day)	0.91	12.5	-18.67

5. Does the edge survive if no opening price is used for trading?

Same gap signal (k=0.5, w=60, h=2). Variant A enters at the CLOSE of the day after the signal day, variant B enters at the close of the signal day itself (not allowed by the brief, which says next open,

shown only to see if the edge is about the opening print). Both trade close to close.

variant	net Sharpe	CAGR %	trades
A: enter close t+1, close-to-close	0.39	4.87	661
B: enter close t, close-to-close (not allowed, for diagnosis)	1.17	18.23	661

6. Alpha and beta of the chosen rule against always-long Nifty (open to open)

average net exposure (long minus short): 0.37 of capital
beta to Nifty: 0.33 (t=4.6)
alpha: 14.7% per year (t=3.02)